

Vydora

Product · Architecture · Business Brief

Investors · Lecturers · Colleagues · Partners

Mobile-first collaborative video editor · Black + gold

Vydora — Product, Architecture & Business Brief

Audience: Investors · Lecturers · Colleagues · Partners
Product: Mobile-first collaborative video editor with AI assist and live team co-editing
Brand: Black + gold (#F5C518) · Domains: `vydora.io` / `vydora.app`
Version of this doc: July 2026 (aligned with current codebase)

1. One-line pitch

Vydora is CapCut for teams — a mobile video editor that combines consumer-grade AI editing with *live collaborative timelines*, guest review links, and role-based project workflows so creators and student/creator teams can go from raw clip to shared export without leaving the phone.

2. Executive summary

Short-form video is how brands, students, creators, and campuses communicate — but the tools people use are split:

Need	Typical tool	Gap
Fast AI edits on phone	CapCut, VN, InShot	Built for <i>solo</i> creators; weak real-time team editing
Team review & approval	Frame.io, Filestage	Desktop-first, expensive, not an editor
Live co-edit	Rare on mobile	Almost none combine editor + live presence + chat
Guest feedback	Email / WhatsApp	No timecode comments, no project context

Vydora closes that gap: one mobile app where you edit like CapCut *and* collaborate like a lightweight Frame.io — with AI tools (captions, silence/filler rough-cut, shorts suggest, beats, auto-reframe) and cloud export baked in.

We are building for:

1. **Creator teams** (2–10 people shipping Reels/TikTok/Shorts weekly)
 2. **Students & campus clubs** (group projects, events, summit intros)
 3. **Small agencies / freelancers** who need client guest review without enterprise software
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3. The problem (why this matters)

1. **Editing is social, tools are solo.** Most CapCut-class apps assume one person owns the timeline. Real projects involve shooters, editors, captioners, and approvers.
2. **Feedback is broken.** Stakeholders reply in WhatsApp with “change the part at 0:42” — no pin on the frame, no version history, no locked final.
3. **AI is bolted on, not team-aware.** Auto-captions help one editor; they don’t help a Viewer request Editor access, or a client leave timestamped notes without an account.
4. **Desktop collab doesn’t travel.** Students and creators work on phones first. Desktop-only review tools don’t match how they already create.

Insight: The next winning editor isn’t “more filters.” It’s **shared editing + AI speed + mobile-native workflow.**

4. Solution — what Vydora is

Vydora is a **mobile-first collaborative video production workspace**:

- **Edit** — CapCut-style timeline, tools, AI assist, publish presets
- **Collaborate** — live presence, cursor/playhead sync, project chat, role upgrades
- **Review** — guest links (no signup) with timestamped comments
- **Ship** — cloud FFmpeg export (720p / 1080p / 4K), version history, Final Locked status
- **Grow** — referral codes → Pro trial; freemium path toward paid plans

Brand personality: dark cinematic UI, gold accent, “make a reel” energy — premium but approachable for students and creators.

5. Product surface (what exists today)

5.1 Core editing

- Multi-clip timeline: split, duplicate, delete, reorder / assemble
- Preview + fullscreen playback
- Beginner-friendly toolbar (Captions, Assemble, Filter, Text, Split) + searchable “More tools” catalog
- Looks: filters, effects, animation, color grade, curves/LUT, canvas, crop, stabilize, auto-reframe
- Text & graphics: captions, title cards (“Sheet”), flyer, stickers, overlay / PiP + masks
- Advanced: keyframes, compound clips, adjustment layers, speed curves, transitions, multi-cam, motion track, rough cut
- Audio: music library + weekly drop, SFX, mixer (duck / denoise / EQ), voiceover, beats
- Publish presets: TikTok / Reels / Shorts / square / YouTube

5.2 AI suite (wired to backend `/api/v1/ai`)

Feature	What it does
Auto captions	Speech → timed segments / words / SRT; speaker-colored turns
Silence detect	FFmpeg-based keep/cut ranges for rough cut
Filler detect	Whisper-backed "um/uh" style hits + keep windows
Shorts suggest	Heuristic hooks for 15–60s cuts
Beat detect	Onset/energy peaks for music-synced edits
Auto-reframe	Motion/edge analysis → 9:16 keyframes
Motion track	Frame analysis for tracking overlays
Multi-cam sync	Audio correlation offset between cameras

Providers: Gemini, Groq Whisper, OpenAI Whisper (optional keys) + local FFmpeg where no key is needed.

5.3 Collaboration

- Project roles: **Owner / Editor / Viewer**
- Email invites + deep links (`vydora://invite`)
- Invite-request approval (Zoom-style) when Editors invite
- **Role upgrade requests** (Viewer → Editor) with live Owner admit
- In-editor **Collaboration Desk**: presence, chat, activity
- Timecode + canvas-pin comments
- **Realtime STOMP**: timeline sync (LWW), live cursors/playheads, comments, messages, presence, user notifications

5.4 Guest review (differentiator)

- Create / list / revoke review links
- Guests watch + leave timestamped comments **without an account**
- Links: `vydora://review/:token` , `https://vydora.io/review/:token`

5.5 Projects, export, versions

- Project CRUD; statuses: Active / Archived / Draft / **FinalLocked**
- Visibility: Private / Team / Public
- Cloud export jobs: MP4 / MOV / WebM × 720p / 1080p / 4K with progress polling
- Version history: snapshot + restore
- Media library; project covers from thumbnails / first frames

5.6 Account, growth, UX

- Email/password + Google Sign-In
- Forgot password → OTP → reset
- Referral hub (`VYD-` codes) → soft Pro / trial

- Pro screen (Paystack WebView scaffold; production billing still being wired)
- Onboarding + **5-minute Wow Path** (seeded first edit → captions → export → invite)
- Dashboard: Get-started cards, explore tools, typing hero, projects with covers
- Settings: theme, auto-save, proxy editing, notifications, presence, export quality

5.7 Honest maturity notes (for lecturers & investors)

Area	Status
Editor + collab + AI APIs + export	Built and integrated
Guest review + referrals	Built
Paystack Pro billing	UI scaffold; not fully production-wired
Horizontal scale (Redis broker / rate limit)	Designed for; single-instance today
Some advanced tools	Depth varies; CapCut-parity is a roadmap, not a claim of 1:1 feature match

6. User journeys (how people use it)

Splash → Sign in / Sign up (Google optional)

→ Onboarding / Wow Path (optional)

→ Tabs: Edit | Projects (exports) | Activity | Me

Typical creator path

New project → Upload → Editor (AI captions / rough cut) → Export → Share

Typical team path

Owner creates project → Invite Editors/Viewers

→ Live co-edit (presence + cursors + chat)

→ Guest review link to client/lecturer

→ Version snapshot → FinalLocked → Export

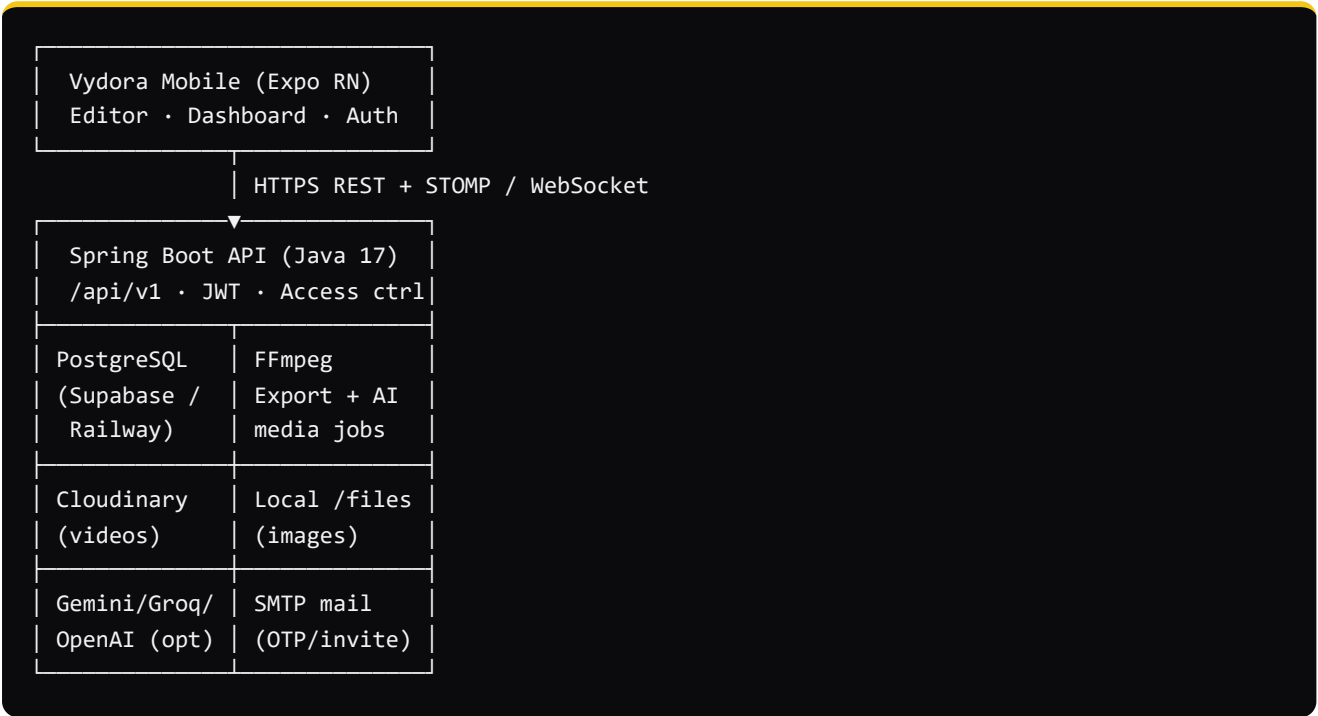
Typical student path

Group assignment → shared project → Viewer comments → request Editor

→ Captions for accessibility → Export for submission

7. System architecture

7.1 High-level



7.2 Frontend stack

Layer	Choice
Runtime	Expo ~54, React Native 0.81, React 19
Navigation	React Navigation (stack + tabs)
Media	expo-video / av / audio / image-picker / thumbnails / file-system
Auth	JWT + refresh; Google via expo-auth-session
Realtime	@stomp/stompjs
Forms	react-hook-form + zod
Brand	Dark #0B0B0D + accent #F5C518

7.3 Backend stack

Layer	Choice
Runtime	Java 17, Spring Boot 3.2
API	REST under <code>/api/v1</code>
DB	PostgreSQL + Flyway migrations; JPA validate
Auth	Stateless JWT, hashed refresh tokens, BCrypt, Google ID tokens
Realtime	STOMP over SockJS <code>/ws</code> + raw <code>/ws-native</code>
Export / AI media	FFmpeg / ffprobe
Video CDN	Cloudinary
Images	Local disk served at <code>/files/**</code>
Deploy	Docker + Railway-oriented prod profile

7.4 Major API domains

`auth` · `projects` · `clips` · `members` · `comments` · `messages` · `notifications` · `versions` · `uploads` · `exports` · `review` · `referrals` · `ai/*` · `health`

Realtime topics include project edits, cursors, comments, messages, presence, and per-user notification queues.

8. Architecture decisions (and why)

These are deliberate choices — useful for lecturers grading design reasoning and for investors assessing engineering maturity.

Decision	Rationale
Mobile-first Expo app	Creators and students already shoot and edit on phones; native reach via one codebase
Separate Spring Boot API	Strong typing, security, long-running FFmpeg jobs, clear ownership vs “all-in-one BaaS”
Flyway-owned schema	Predictable migrations on Postgres/Supabase; Hibernate validates only
Hybrid storage	Cloudinary for durable video CDN; local files for cheap avatars/covers (LAN-friendly for Expo Go)
Client timeline as export source of truth	Bake what the editor actually shows (timeline JSON) rather than reconstructing from fragmented tables
Dual AI strategy	Cloud Whisper/Gemini for speech; local FFmpeg for silence/beats/reframe/multicam (works offline-of-keys, lower cost)
STOMP + SimpleBroker	Fast iteration for collab; explicitly noted that Redis is next for multi-instance scale
Guest review tokens	Async stakeholder feedback without forcing signup friction
Roles + upgrade requests	Mirrors real classroom/agency permission dynamics
Long-lived mobile JWTs (dev-friendly)	Editor sessions shouldn't expire mid-cut; refresh + revoke on logout/reset
Rate-limit auth endpoints	Basic abuse protection; Redis planned for multi-node
Wow Path + referrals	Product-led growth before heavy paid acquisition

9. Competitive landscape & edge

9.1 Comparison snapshot

Capability	CapCut	VN / InShot	Frame.io	
Mobile AI editing	★★★★★	★★★★★	★	★★★★★
Live multi-user timeline	★	★	★★	★★★★★
In-editor chat + presence	★	★	★★★	★★★★★
Guest review (no account)	★	★	★★★★★	★★★★★
Role upgrade workflow	★	★	★★	★★★★★
Cloud team export + versions	★★	★	★★★★★	★★★
Price accessibility (students)	Free tier	Free	Enterprise-ish	Freemium + referral Pro
Conference / flyer / sheet tools	Partial	Partial	N/A	Designed-in

9.2 Why we can win a *segment* (not “beat CapCut globally”)

We do **not** claim to out-feature CapCut’s entire effect library on day one. We win where CapCut is weak:

1. **Team-native editing** — live cursors, LWW timeline sync, presence, chat inside the editor
2. **Review without friction** — guest links for lecturers, clients, club executives
3. **Permission realism** — Viewer → Editor upgrade requests; FinalLocked handoff
4. **Campus / Africa-friendly GTM** — referral Pro, mobile-first, Paystack-oriented billing path
5. **Vertical tools** — conference slates, flyer, sheet / title cards for events and summits
6. **Guided first success** — Wow Path so first-time users ship a reel in minutes

Positioning line:

CapCut makes *you* faster. Vydora makes *your team* faster.

9.3 Moat trajectory

Near-term	Medium-term
Collab UX + guest review habit	Org/workspace accounts for clubs & agencies
Template + music packs	Marketplace + brand kits
Referral graph	Campus ambassador program
Export quality	Proxy editing + cloud render tiers

10. Intended business model

10.1 Monetization

Tier	Who	What they get	How we charge
Free	Individuals / students	Core edit, limited AI / export quality, watermark options as needed	\$0
Pro	Serious creators	Higher export quality, more AI runs, templates, no/low watermark	Monthly / yearly (Paystack)
Team (planned)	Clubs, agencies	Seats, shared brand kit, admin, priority export	Per seat / month
Education (planned)	Lecturers / departments	Classroom workspaces, guest review quotas	Annual campus license

Growth loops already in product:

- Referral codes → Pro trial (`pro_until`)
- Wow Path → export → invite teammate
- Guest review → guest becomes registered user

10.2 Unit economics (directional)

- **Costs:** Cloudinary storage/bandwidth, Whisper/Gemini API calls, FFmpeg CPU on export, Postgres hosting
- **Levers:** Prefer FFmpeg-local AI where possible; cache captions; tier AI/export quotas on Free vs Pro
- **ARPU path:** Free → Pro via referral or export friction; Team when projects regularly exceed 2 members

10.3 Go-to-market

1. **Campus-first:** film clubs, media students, hackathons, entrepreneurship courses
2. **Creator pods:** 3–5 person TikTok/Reel teams
3. **Event vertical:** summit intros, conference lower-thirds, end cards (Flyer / Sheet tools)
4. **Content:** “edit with your team on phone” demos vs CapCut solo workflow

11. Market context (for investors)

- Short-form video remains the default content format for Gen Z / Gen Alpha communication
- Creator economy tools expand from *solo influence* → *small production teams*
- Collaboration SaaS (Notion, Figma, Frame.io) proved willingness to pay for multiplayer creative work
- Mobile-first markets (Africa, SEA, LatAm) often skip desktop-heavy creative suites

Vydora’s bet: multiplayer editing on mobile is underserved; AI makes the editor “good enough” while collab becomes the reason to switch.

12. Academic / lecturer framing

12.1 What this project demonstrates

Course theme	How Vydora maps
Software engineering	Full-stack product: RN client + Spring API + DB + realtime + media pipeline
Distributed systems lite	STOMP presence, LWW timeline sync, async export worker
HCI / UX	Beginner toolbar, Wow Path, guest review without signup
AI systems	Hybrid cloud LLM/ASR + classical media DSP (FFmpeg)
Security	JWT, refresh hashing, role-based access, public tokenized review routes
Entrepreneurship	Freemium + referral, competitive positioning, GTM

12.2 Suggested demo script (5–7 minutes)

1. Create project → upload clip
2. Auto captions → silence/filler rough cut
3. Invite second user → show live cursor / chat
4. Open guest review link on another device → leave comment

5. Snapshot version → export 1080p → share

12.3 Learning outcomes for teammates

- How product requirements become API modules
- Why hybrid storage and export-as-job matter
- Tradeoffs: SimpleBroker vs Redis; Free AI vs paid ASR
- How to pitch *segment leadership* instead of “we beat CapCut at everything”

13. Risks & mitigations

Risk	Mitigation
CapCut feature gravity	Differentiate on collab + review; don't compete on effect count alone
AI cost spikes	FFmpeg-local tools; quota Free/Pro; cache transcripts
Single-instance realtime	Redis broker + sticky sessions on roadmap
Export CPU cost	Queue, resolution tiers, Pro priority
Billing incomplete	Finish Paystack; keep referral Pro for early traction
Content moderation	Report flows + ToS; private-by-default projects

14. Roadmap (indicative)

Now — polish & reliability

- Harden export bake parity with editor timeline
- Production Paystack Pro
- Collab conflict UX + reconnect resilience

Next — team product

- Workspaces / brand kits
- Proxy editing for large files
- Education guest-review quotas

Later — platform

- Template marketplace
- Org SSO / campus licenses
- Multi-region render workers

15. The ask (customize per audience)

Investors

- Capital for: cloud/AI burn, campus GTM, 2–3 engineers focused on collab + export reliability
- Traction plan: campus cohorts → creator pods → Team SKU

Lecturers

- Recognition of full-stack + product thinking, not only UI screens
- Feedback on evaluation criteria: architecture quality, UX research, business clarity

Colleagues

- Shared north star: **team-native mobile editing**
 - Ownership map: Editor UX · Collab/realtime · AI/export · Growth/billing · Brand/dashboard
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16. Appendix — module map

Mobile (`vydora/`)

Dashboard · Editor · Auth · Projects · Invite · Guest Review · Export · Referral · Pro · Settings · Activity · Wow Path · AI service clients · STOMP sync

Backend (`vydora-backend/`)

auth project clip member comment message notification version upload export ai review
referral mail ws security config

17. Closing statement

Vydora exists because **video creation became a team sport**, while the most popular mobile editors stayed solo apps. We combine CapCut-like AI speed with live collaboration, guest review, and a freemium growth model designed for creators and campuses.

Edit alone if you want. Ship together if you want to win.

Document generated from the Vydora application and API codebases for internal pitching and academic presentation. Update billing/roadmap sections as product status changes.